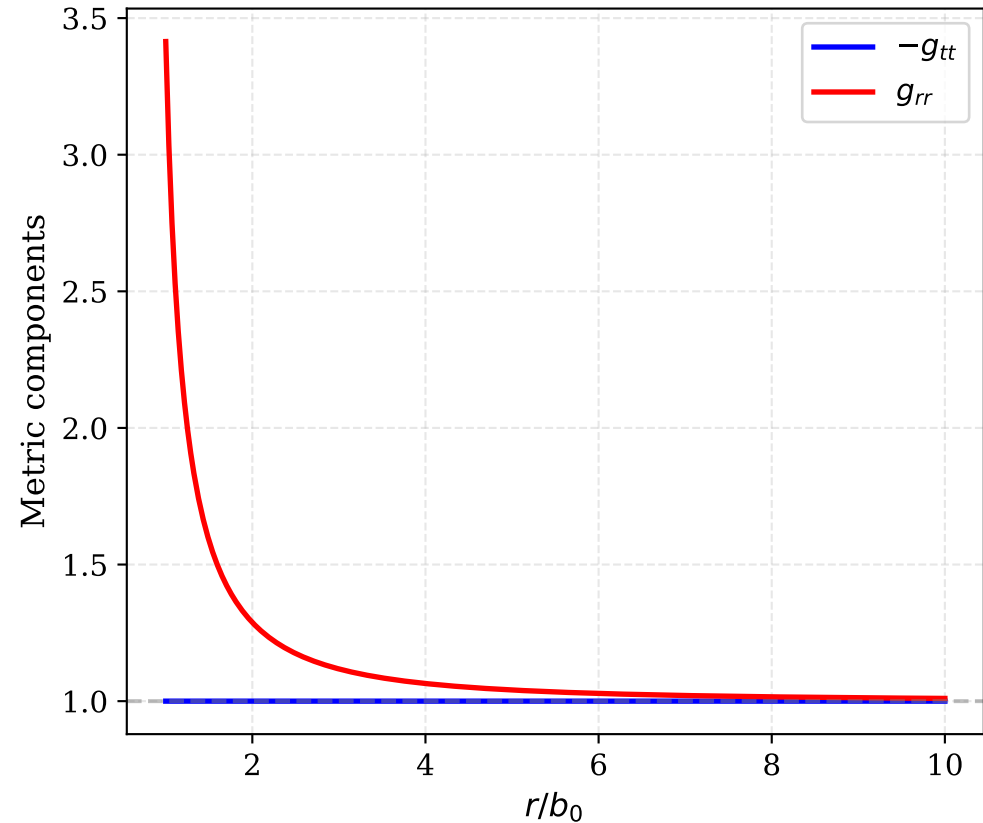
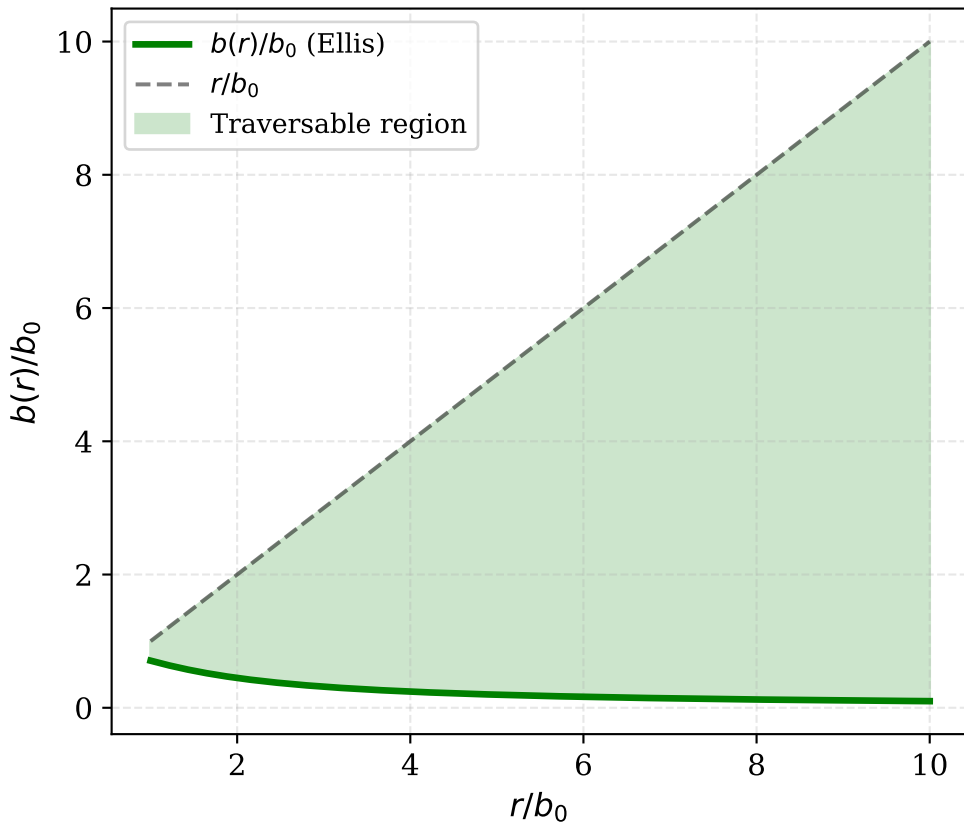


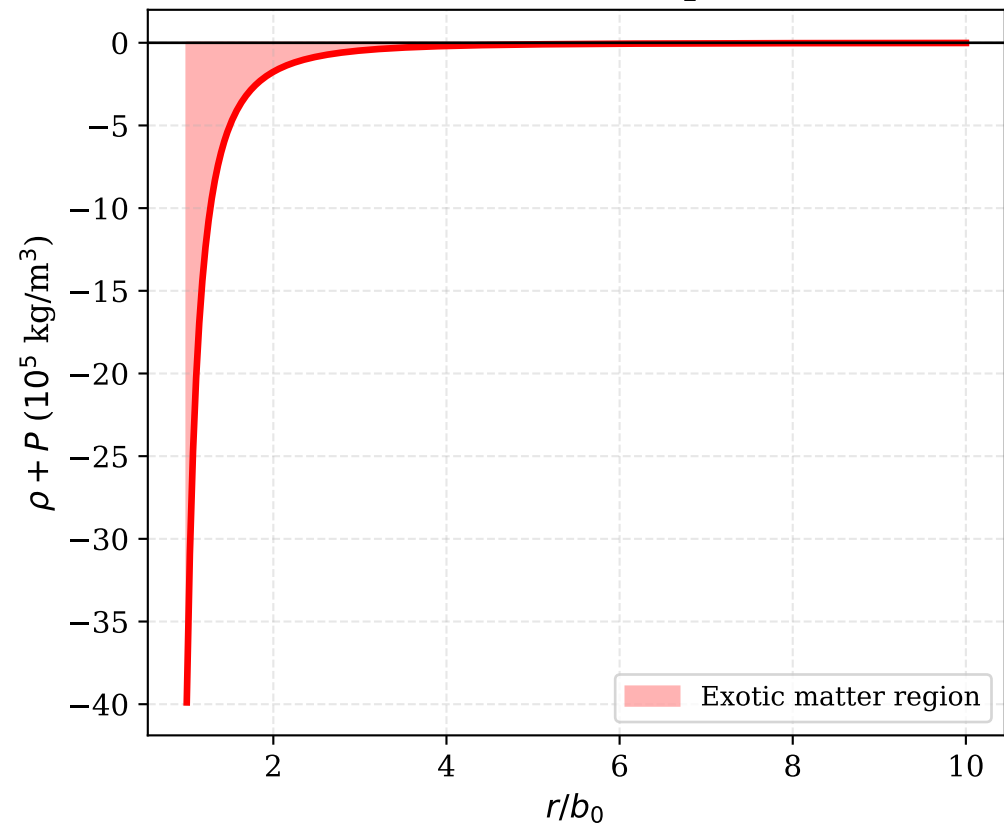
(a) Metric Tensor Components



(b) Throat Function (Flare-Out)



(c) Exotic Matter Requirement



Morris-Thorne Traversability Criteria

Test Case: $b_0 = 1$ km, $r = 2b_0 = 2$ km

Metric Components:

- $g_{tt} = -1.000$ ✓
- $g_{rr} = 1.288$ (> 1 outside throat) ✓

Traversability Conditions:

- Outside throat: $r > b(r)$
 $2000 \text{ m} > 1553 \text{ m}$ ✓
- Flare-out: $db/dr < 1$
 $0.603 < 1$ ✓
- Exotic matter: $\rho + P < 0$
 $-1.75 \times 10^5 \text{ kg/m}^3 < 0$ ✓

Result: TRAVERSABLE (all criteria satisfied)

Note: Standard UQFF vacuum has $\rho + P = +7.32 \times 10^7 \text{ kg/m}^3$ (too positive). Requires negative pressure component.